

JPE Replication Report

PUBLISHED

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1 Paper Identification

ID	20240098
Author	Sollaci
Title	Managers and Productivity in Retail
Iteration	1

Some useful information up front:

- [JPE Data and Code Availability Policy](#)
- [Step by step guidance from Data Editor for Authors](#)
- [Template README](#)

2 Summary

In assessing compliance with our [Data and Code Availability Policy](#), our reproducibility team has identified the following issues, which we ask you to address:

1. In your readme, please include a statement of rights (see below)
2. Please add more details on computational requirements - we need version numbers for all used libraries, your used hardware, OS, and so on. Currently, there is no list of required libraries. This holds for python, stata and R.
3. If there is a random component in your computations, please set a seed and document this in your readme.
4. There are several required changes to the readme below.
5. We have some difficulty mapping outputs to paper exhibits (section 9.3 below). Table 2 seems to be off by 1.
6. In general, there seem to be small divergence across our results and yours, could be due to a missing seed somewhere or out of data output in your paper - please check.

3 General

☐ Data Availability and Provenance Statements

- ☐ Statement about Rights - Part 1: Right to use the data (e.g. *did the authors lawfully obtain the data*)
- ☐ Statement about Rights - Part 2: Right to publish the data (e.g. *are human subjects involved and did permission been granted to publish their data?*)
- ☐ License for Data (optional, but recommended)
- ☐ Details on each Data Source
- ☐ Dataset list
- ☐ Computational requirements
 - ☐ Software Requirements
 - ☐ Controlled Randomness (as necessary)
 - ☐ Memory, Runtime, and Storage Requirements
- ☒ Description of programs/code
 - ☐ License for Code (Optional, but recommended)
- ☐ {REPLICATOR} to Replicators
 - ☐ Details (as necessary)
- ☐ List of tables and programs
- ☐ References (Optional, but recommended)

4 High-level Description of Research Project

This research project...

- ☐ uses observational data.
- ☒ uses *confidential* observational data.
- ☐ uses data generated via experiment or trial by the authors (in field or lab).
 - ☐ For lab experiments: The code to implement the experimental protocol (z-Tree, o-Tree etc) is included in the package.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF document outlining the design of the experiment, including information on the selection and eligibility of participants is included.
 - ☐ A PDF copy of the instructions given to participants in both original language and an English translation is included.
- ☐ uses data generated via survey by the authors (in field or online)
 - ☐ The programs used to administer the survey (e.g. qualtrics survey file) are included.
 - ☐ A PDF copy of IRB approval is contained in the package.
 - ☐ A PDF of the original questionnaire(s) and information on the selection and eligibility of participants is included.
- ☒ is a simulation study: the data are generated by an algorithm. (Parts of it.)
- ☐ is a numerical illustration: no data are used but code is used to produce illustrative output (tables, figures)

5 Data description

For any data that cannot be provided as part of the replication package, please provide the following information as part of the README.

{REQUIRED} Please specify how long the data will be preserved in the restricted-access location.

{REQUIRED} Please provide affirmation of support for replication checks. – As per the [JPE policy](#), “In cases where data cannot be published in an openly accessible trusted data repository like the JPE dataverse, authors who have requested an exemption to publish them at the time of first submission must commit to preserving the data and code for a period of no less than five years following the publication of the paper. They should also provide reasonable assistance to requests for clarification and replication.”

{REQUIRED} Please provide a codebook for the data. – The codebook should at a minimum describe the variables obtained from the data source, in a manner that will let others verify that they have obtained a substantially similar dataset if successfully obtaining access to the data.

5.1 Data Sources

file/name	public	private	DOI/URL	Access		
				described	cited readme	cited paper
data/A_dataset.dta	no	yes	no	no	no	no
data/B_dataset.dta	no	yes	no	no	no	no

5.1.1 Data

Data are not cited, no access description provided either, no information regarding the data is provided.

- ☐ Dataset is provided in public deposit
- ☒ Dataset is PRIVATELY provided (not for publication)
- ☐ DOI or URL is provided, and works: <http://data.bls.gov/cgi-bin/surveymost?sm+08>
- ☐ Access conditions are described:
 - ☐ No information on access conditions.
- ☐ Data are not cited, but a working paper, article, or other document is cited (not a data citation)
- ☐ Data are cited
 - ☐ in the manuscript
 - ☐ in the README:

6 Requirements

- ☒ README is in TXT, MD, or PDF format

- ☒ README is accessible at the root of the package (not inside a folder)
- ☐ Title conforms to guidance (starts with "Data and Code for: Title of Paper" or "Code for: Title of Paper", is properly capitalized)
- ☐ Authors (with affiliations) are listed in the same order as on the paper

{REQUIRED} Please review the title of the deposit as per our guidelines.

{REQUIRED} Please review authors and affiliations on the deposit. In general, they are the same, and in the same order, as for the manuscript; however, authors can deviate from that order.

6.1 Stated Requirements

- ☐ No requirements specified
- ☒ Software Requirements specified as follows:
 - R, no version specified
 - Stata, no version specified
- ☒ Computational Requirements specified as follows:
 - 'reghdfe', 'did_imputation', and 'event_plot' packages (stata)
 - ebayes.ado and fese_fast.ado (provided in repository)
- ☐ Time Requirements specified as follows:
 - Length of necessary computation (hours, weeks, etc.)
- ☐ Requirements are complete.

For missing requirements, see the list of required changes in [Section 9](#)

7 Analyzing Data and Code Files

7.1 Potential Personal Identifiable Information (PII)

⚠ We found the following instances of potentially personally identifying information. This may be completely legitimate but might be worth checking. *As a reminder, privacy legislation in many countries (e.g. GDPR in EU) prohibits the dissemination of personal identifiable information without prior (and documented) consent of individuals.* If indeed you want to publish such information with your replication package, you should probably have obtained IRB approval for this - please check!

Summary: - Data files with PII indicators: 2 - Variables flagged in data: 44 - Code files with PII references: 19 - PII references in code: 382

7.1.1 Summary of Flagged Files

File Type	File	Variables/References	PII Categories
Data	A_dataset.dta	8	loc, location, name, school, degree, city
Data	B_dataset.dta	36	lat, name, address, coord, lon, country, minute, loc, location, city
Code	A_estimate_fe.R	14	name, network, lat
Code	A_estimate_gfe.R	12	name, network, lat
Code	A_event_study.do	64	loc, name, lon, son, lat
Code	A_fixed_effects.do	10	name, lat, loc, location, lon
Code	A_summary_stats.do	10	loc, location, lat, name
Code	B_estimate_fe.R	14	name, network, lat
Code	B_estimate_gfe.R	12	name, network, lat
Code	B_event_study.do	64	loc, name, lon, son, lat
Code	B_fixed_effects.do	6	name, lat, loc, location
Code	B_gender_moves.do	9	gender, lon, loc, location
Code	B_summary_stats.do	11	name, lat
Code	calibrate_NAM.py	1	lat
Code	ebayes.ado	52	name, loc, lat, second, degree, lon
Code	ebayes.ado	52	name, loc, lat, second, degree, lon
Code	fese_fast.ado	18	lat, city, name, loc
Code	fese_fast.ado	18	lat, city, name, loc

File Type	File	Variables/References	PII Categories
Code	<code>run_stata_codes.do</code>	1	gender
Code	<code>selection_based_NAM.do</code>	6	lat
Code	<code>simulate_selection.py</code>	8	name, lat, loc

See [Appendix](#) for detailed listing of all flagged instances.

7.1.2 File Paths Report

Generated on 2026-04-02T12:30:03.688

Warning: Our search on file path types is imperfect and incurs both type 1 and type 2 errors. We aim to strike a reasonable balance between both. The below table is therefore only indicative. Detailed listings can be found in the appendix to this report.

In this table we analyse all files which contain source code (not data and documentation), and we report the grand total of each type of filepath we encountered.

Please check and replace any `windows` filepaths with unix compliant paths, insofar as this is possible in your setup. (Notice that `STATA` allows `/` to be a filepath separator on a windows platform, hence this is a requirement for all `STATA` applications.)

Files Analyzed	Windows Paths	Unix Paths	Mixed Paths	Drive Letters (C:\ etc)
24	4	7	2	0

7.1.3 Duplicate Files Report

We found the following duplicate files:

Filename	Size (MB)	Checksum (MD5)
/Replication/CodeB/Stata codes/fese_fast.ado	0.0	e18776996ac50b147dbd51ba774dc0eb8fe2da6f
/Replication/CodeB/Stata codes/ebayes.ado	0.01	295ddb30a99b2a2c61ecc67819429559ff3444c6

7.1.4 Zero Size Files Report

We did not find any zero sized files.

7.1.5 Large Files Report

We did not find any files larger than 100MB.

7.2 Code description

- Folder A contains 2 R scripts, a master Stata do-file which references to 5 separate stata files. Additionally it contains 2 python files and another separate stata do-file for the simulation part of the study. Folder B replicates the same structure of folder A for the analysis for company B instead of company A, excluding the simulation.
- It is unclear from the ReadMe which in-text numbers are replicated where. It would be very helpful if the authors provided more details. For instance, "the numbers cited in the related discussion in the text" is not precise enough for the replicator to associate the code with specific numbers in the text.

cloc github.com/AIDanial/cloc v 2.02 T=0.04 s (538.3 files/s, 116569.6 lines/s)				
Language	files	blank	comment	code
Stata	16	1049	273	2561
R	4	196	80	380
Python	2	76	48	170
Markdown	1	110	0	38
---	---	---	---	---
SUM:	23	1431	401	3149

- ☐ The replication package contains a "main" or "master" file(s) which calls all other auxiliary programs.

{SUGGESTED} We strongly advise the use of a single (or a small number of) main control file(s) to automatically reproduce all figures and tables in the paper, without manual interaction.

NOTE: In-text numbers that reference numbers in tables do not need to be listed. Only in-text numbers that correspond to no table or figure need to be listed.

7.3 Computing Environment of the Replicator

- Nuvolos on an Mac Laptop M1 MacBook Air, MacOS Tahoe 26.3, 8 GB of memory, (M1 Processor, 8 cores)
- R 4.5.2

- Stata/MP 18.0
- JupyterLab 4.5.4 with Python 3.13 and Julia 1.12.4

8 Replication

- Downloaded code and data from dropbox.
- Committed code to git repo.
- Installed the doBy, and lfe packages in R, (though it is not detailed in the ReadMe that one needs to install it for the code to run).
- Ran code in CodeA folder R-scripts as per README, for company A, the code ran through.
- Ran Stata code through the master do-file as per ReadMe, but the code broke:

```
.set scheme white_tableau
scheme white_tableau not found r(111);
```

- I turned to running the individual do-files the master calls upon. Starting with stata file A_summary_stats.do but code broke for Table B.1:

```
command grcl1leg is unrecognized r(199);
```

- Ran Stata file A_fixed_effects.do and had to fix "" in path lines, code broke due to "ftools", "require" packages not being installed. After installing these packages the code broke with the following error message:

```
command binscatter is unrecognized r(199);
```

- Ran Stata do file A_group_fixed_effects.do and the same error message as above broke the code.
- Stata do file A_event_study.do ran to completion.
- Stata do file A_counterfactuals ran to completion.
- Moving on to the simulation folder, I amended the paths in the selection_based_NAM.do file after which it ran to completion.
- Moving to the Python scripts which I run in JupyterLab, the calibrate_NAM.py file runs to completion. The simulate_selection.py file broke of due to the paths not being correctly specified. Concretely I changed the set directory lines in the following way:

```
### Set directory

os.chdir("/files/JPE-Sollaci-20240098/replication-package/Replication/CodeA/NAM Simulation")
script_dir = os.getcwd()
parent_dir = os.path.dirname(os.path.dirname(script_dir))
figures_dir = os.path.join(parent_dir, "Results", "Figures")
```


- Next, I move on to company B, in the CodeB folder. The R scripts run to completion. The master do file for the stata code has the same break as for company A. The same goes for the B_summary_stats which breaks for the same reasons detailed above for company A.
- The B_fixed_effects.do and B_group_fixed_effects.do broke for the same reason the A_fixed_effects.do broke.
- B_event_study.do ran to completion.
- B_counterfactuals.do ran to completion.
- B_gender_moves.do ran to completion.

9 Findings

9.1 Missing Requirements

- ☐ Software Requirements
 - ☐ Stata
 - ☐ Version (missing)
 - ☐ R
 - ☐ Version (missing)
 - doBy
 - lfe
 - ☐ Python
 - ☐ Version (missing)
- ☐ Computational Requirements specified as follows:
 - Cluster size, disk size, memory size, etc. (not specified)
- ☐ Time Requirements
 - Length of necessary computation (hours, weeks, etc.) (not specified)

[REQUIRED] Please amend README to contain complete requirements.

You can copy the section above, amended if necessary.

9.2 Data Preparation Code

There are no files that separately prepare the data, the data preparation is handled inside the same scripts as the analysis is done.

9.3 Tables and Figures

- Table 1: Table 1: The ReadMe states that the information displayed in table 1 is saved in 8 different files in the results folder. While these files are created running the R code for both company A and company B, the information displayed in these files does not match the information displayed in table 1. The result below in the screenshot is supposed to correspond with line 2 of table 1.

```
Call:
felm(formula = log_prod_time ~ 1 | manager_id + store_id, data = df_sub,
      cmethod = "reghdfe")

Residuals:
    Min       10   Median       30      Max
-0.32641 -0.06629 -0.00498  0.05850  0.55220

Coefficients:
(No coefficients)

=====### Total Variance ###=====
[1] 0.03717122

=====### Covariance Matrix ###=====
      manager_id store_id
manager_id 0.03813754 -0.04320112
store_id   -0.04320112 0.06682680
attr(,"bias")
      manager_id store_id
manager_id 0.008592190 -0.006272917
store_id   -0.006272917  0.008486226

=====### Covariance - Biased ###=====
      manager_id store_id
manager_id 0.04672973 -0.04947404
store_id   -0.04947404  0.07531303

=====### Correlation ###=====
      manager_id store_id
manager_id 1.0000000 -0.8557429
store_id   -0.8557429  1.0000000

=====### Correlation - Biased ###=====
      manager_id store_id
manager_id 1.0000000 -0.8339608
store_id   -0.8339608  1.0000000

=====### Variance Share ###=====
      manager_id store_id
manager_id 1.025997 -1.16222
store_id   -1.16222  1.79781
```

istent estimate for the relationship between manager and store quality. In practice, this process can involve computing inverses of impractically large matrices, so we employ the method developed by [Gaure \(2014\)](#), again through the `lfe` package in R.

Table 1: Decomposition of the Variance of Productivity

CS Rank	Obs	Company A		
		$\frac{\text{Var}(\mu_i)}{\text{Var}(\log(\text{prod}_{s,i}) - \mu_i)}$	$\frac{\text{Var}(\mu_m)}{\text{Var}(\log(\text{prod}_{s,i}) - \mu_i)}$	$\frac{\text{Cov}(\mu_{s,i}, \mu_m)}{\text{Var}(\log(\text{prod}_{s,i}) - \mu_i)}$
1	215	0.72	0.05	-0.07
2	191	1.84	1.03	-1.13
3	171	0.21	0.12	0.15
4	142	0.48	0.07	-0.02

CS Rank	Obs	Company B		
		$\frac{\text{Var}(\mu_i)}{\text{Var}(\log(\text{prod}_{s,i}) - \mu_i)}$	$\frac{\text{Var}(\mu_m)}{\text{Var}(\log(\text{prod}_{s,i}) - \mu_i)}$	$\frac{\text{Cov}(\mu_{s,i}, \mu_m)}{\text{Var}(\log(\text{prod}_{s,i}) - \mu_i)}$
1	3,710	1.64	1.50	-1.22
2	1,979	0.72	0.22	-0.13
3	1,179	1.04	0.77	-0.62
4	1,008	0.39	0.29	-0.03

rank is the rank of the connected set by size, and *Obs* indicates the number of observations (store*month) in the connected set. Columns 3-5 in this table show the ratio of the variance in the store FE (or manager FE, or their covariance) and the variance in total productivity according to equation (2). Results are computed within the fixed effects connected sets for each company and all estimates adjust for limited mobility bias.

After adjusting for limited mobility bias, Table 1 shows the ratio of the store fixed effects (FE) variance, manager FE variance, and their covariance to the variance in productivity.

Output on the left should correspond with line 2 of manuscript table 1 on the right

- Table 2: The numbers reproduced do not match the table in the manuscript. See screenshot below:

	manager...	store_bin	freq	lab_mana...	lab_store
1	1	1	33	Top	Top
2	2	1	49	Middle	Top
3	3	1	88	Bottom	Top
4	1	2	51	Top	Middle
5	2	2	67	Middle	Middle
6	3	2	76	Bottom	Middle
7	1	3	97	Top	Bottom
8	2	3	69	Middle	Bottom
9	3	3	64	Bottom	Bottom

Reproduced output should correspond to the company A side of table 2

	manager...	store_bin	freq	lab_mana...	lab_store
1	1	1	116	Top	Top
2	2	1	229	Middle	Top
3	3	1	461	Bottom	Top
4	1	2	228	Top	Middle
5	2	2	332	Middle	Middle
6	3	2	285	Bottom	Middle
7	1	3	457	Top	Bottom
8	2	3	322	Middle	Bottom
9	3	3	173	Bottom	Bottom

Reproduced output should correspond to the company B side of table 2

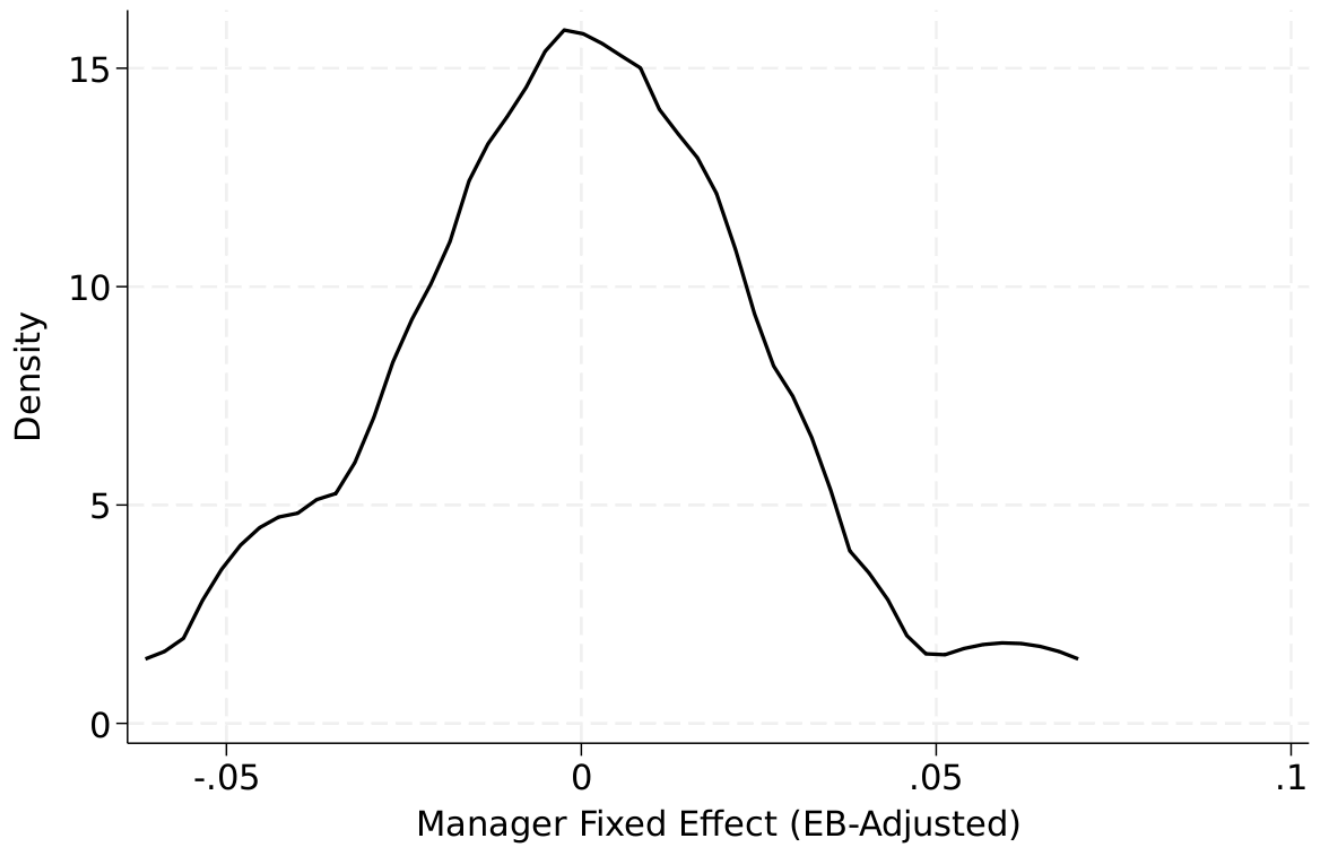
Table 2: Number of manager and store matches

		Stores					
		Company A			Company B		
		Top	Middle	Bottom	Top	Middle	Bottom
Managers	Top	34	45	91	115	242	454
	Middle	50	70	74	225	329	286
	Bottom	98	69	63	459	316	177

Paper version

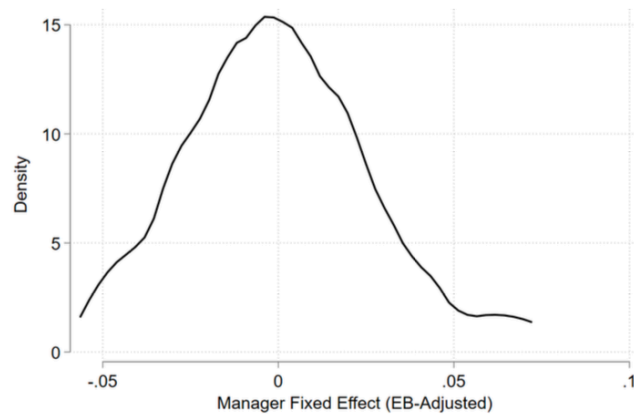
Figure 1

- Figure 2: looks the same.
- Figure 3: Reproduced but it looks different (I include (a) as an example though the others look different as well:

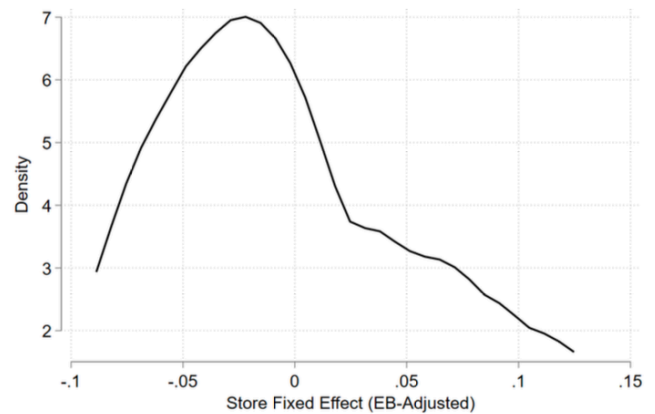


Reproduced output for a)

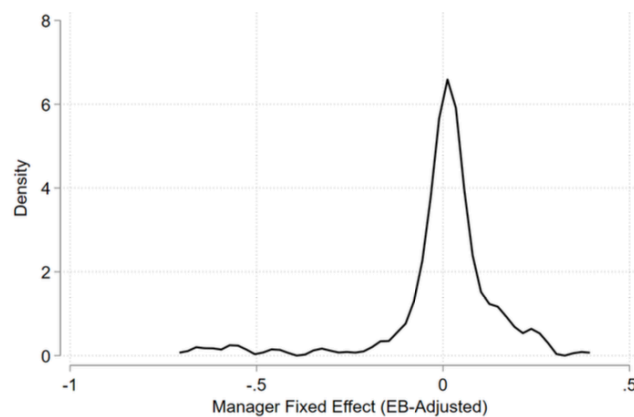
Figure 3: Distribution of Manager and Store Quality (EB-Adjusted)



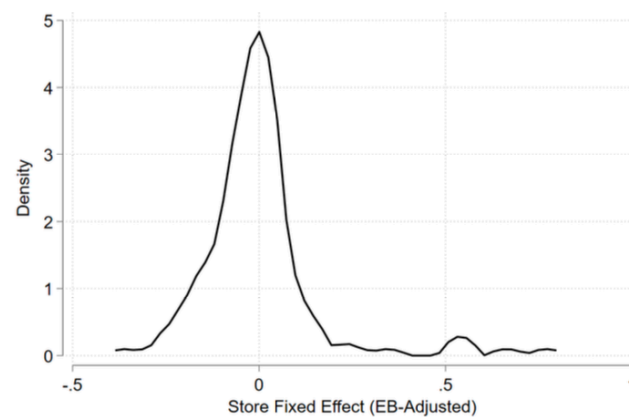
(a) Manager FE - Company A



(b) Store FE - Company A



(c) Manager FE - Company B



(d) Store FE - Company B

Paper version

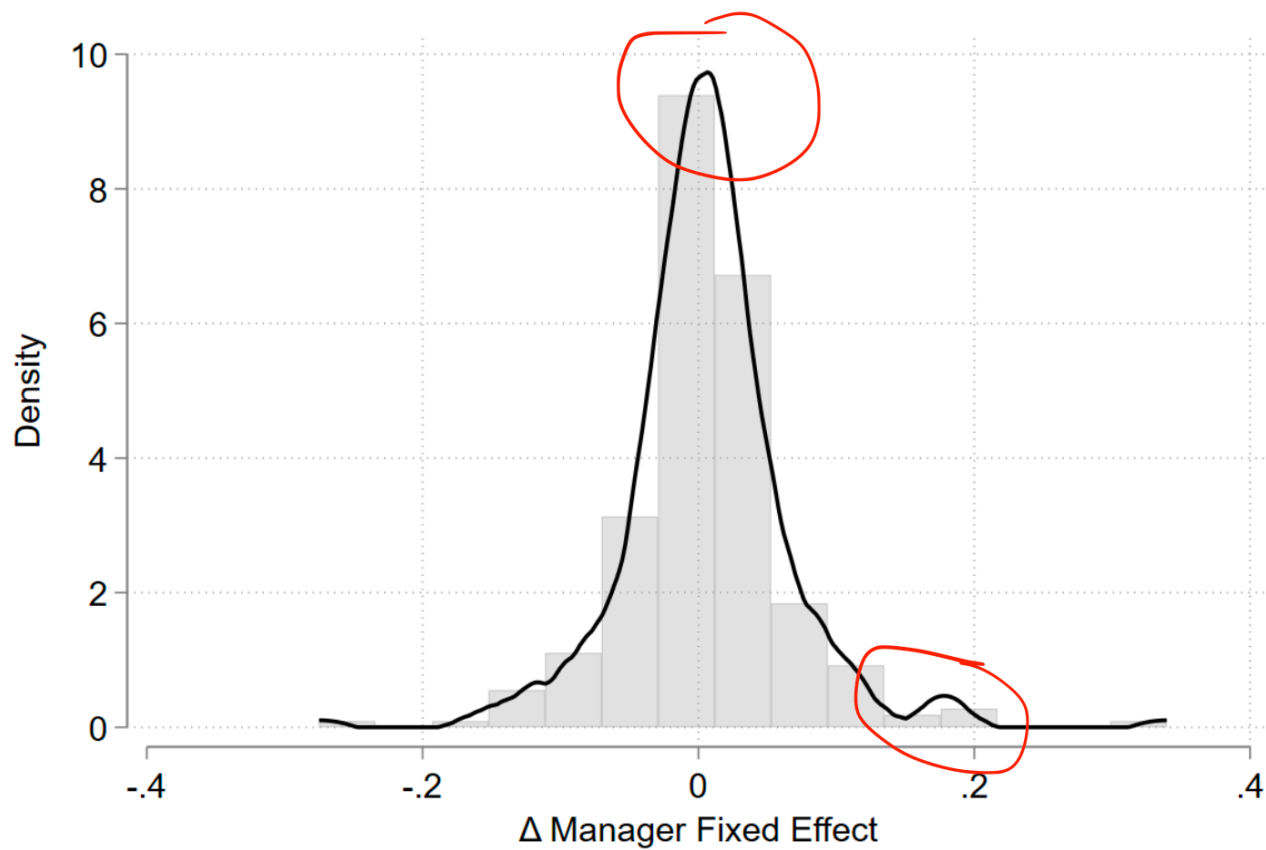
Figure 2

- Figure 4: not reproduced due to code breaking off (see above).
- Figure 5: looks the same.
- Figure 6: looks the same.
- Figure 7: Reproduced, but with differences:



Reproduced output for a)

Figure 7: Distribution of Cha



(a) Manager FE (Company A)

Paper version

Figure 3

- Table 3: Numbers diverge:

start_in[1]		1						
	start_in	move_to	freq	lab_start	lab_go	total_mng	pct	
1	1	1	65	Top	Top	100	65	
2	1	2	18	Top	Middle	100	18	
3	1	3	17	Top	Bottom	100	17	
4	2	1	27	Middle	Top	119	22.68908	
5	2	2	73	Middle	Middle	119	61.34454	
6	2	3	19	Middle	Bottom	119	15.96639	
7	3	1	23	Bottom	Top	165	13.93939	
8	3	2	33	Bottom	Middle	165	20	
9	3	3	109	Bottom	Bottom	165	66.06061	

Reproduced output for company A

start_in[1]		1						
	start_in	move_to	freq	lab_start	lab_go	total_mng	pct	
1	1	1	339	Top	Top	492	68.90244	
2	1	2	94	Top	Middle	492	19.10569	
3	1	3	59	Top	Bottom	492	11.99187	
4	2	1	80	Middle	Top	506	15.81028	
5	2	2	316	Middle	Middle	506	62.45059	
6	2	3	110	Middle	Bottom	506	21.73913	
7	3	1	79	Bottom	Top	603	13.10116	
8	3	2	111	Bottom	Middle	603	18.40796	
9	3	3	413	Bottom	Bottom	603	68.49088	

Reproduced output for company B

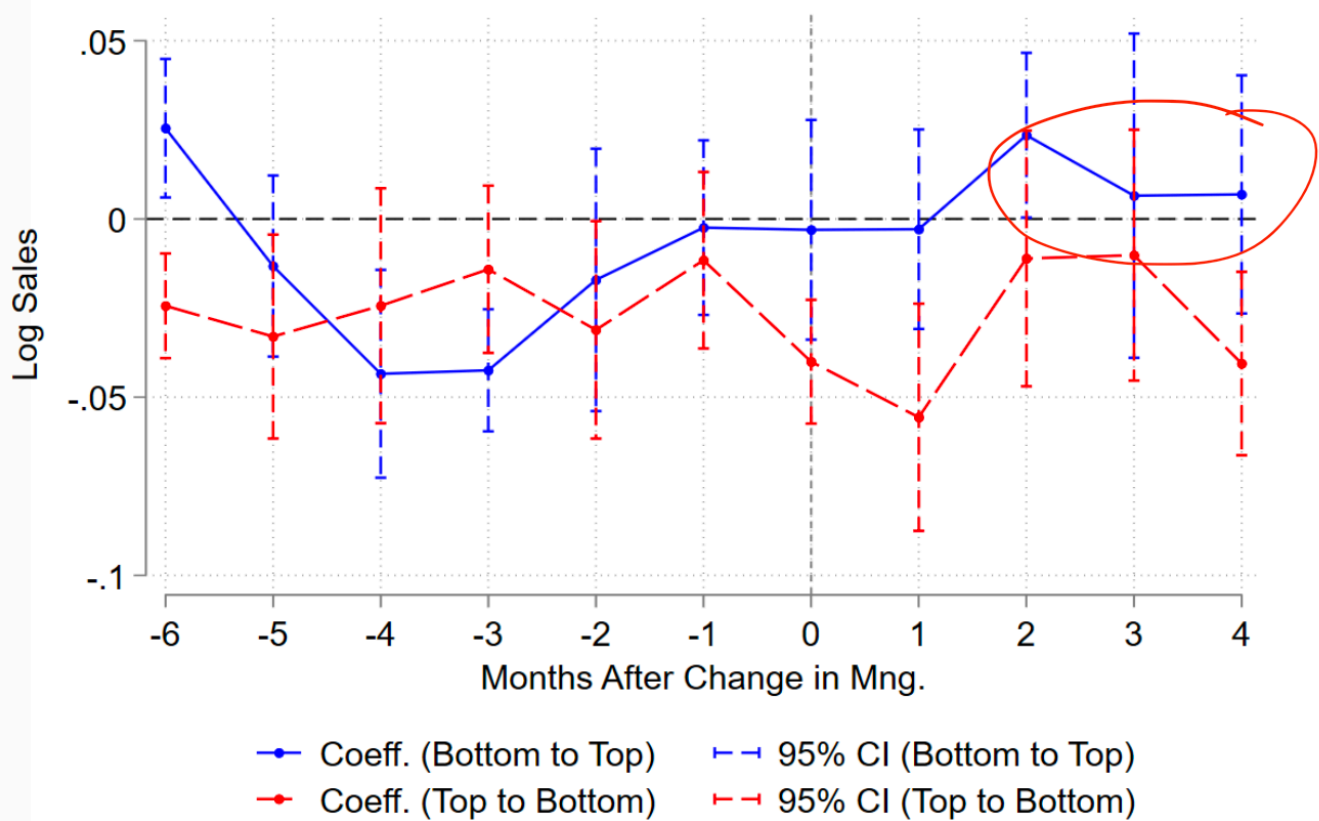
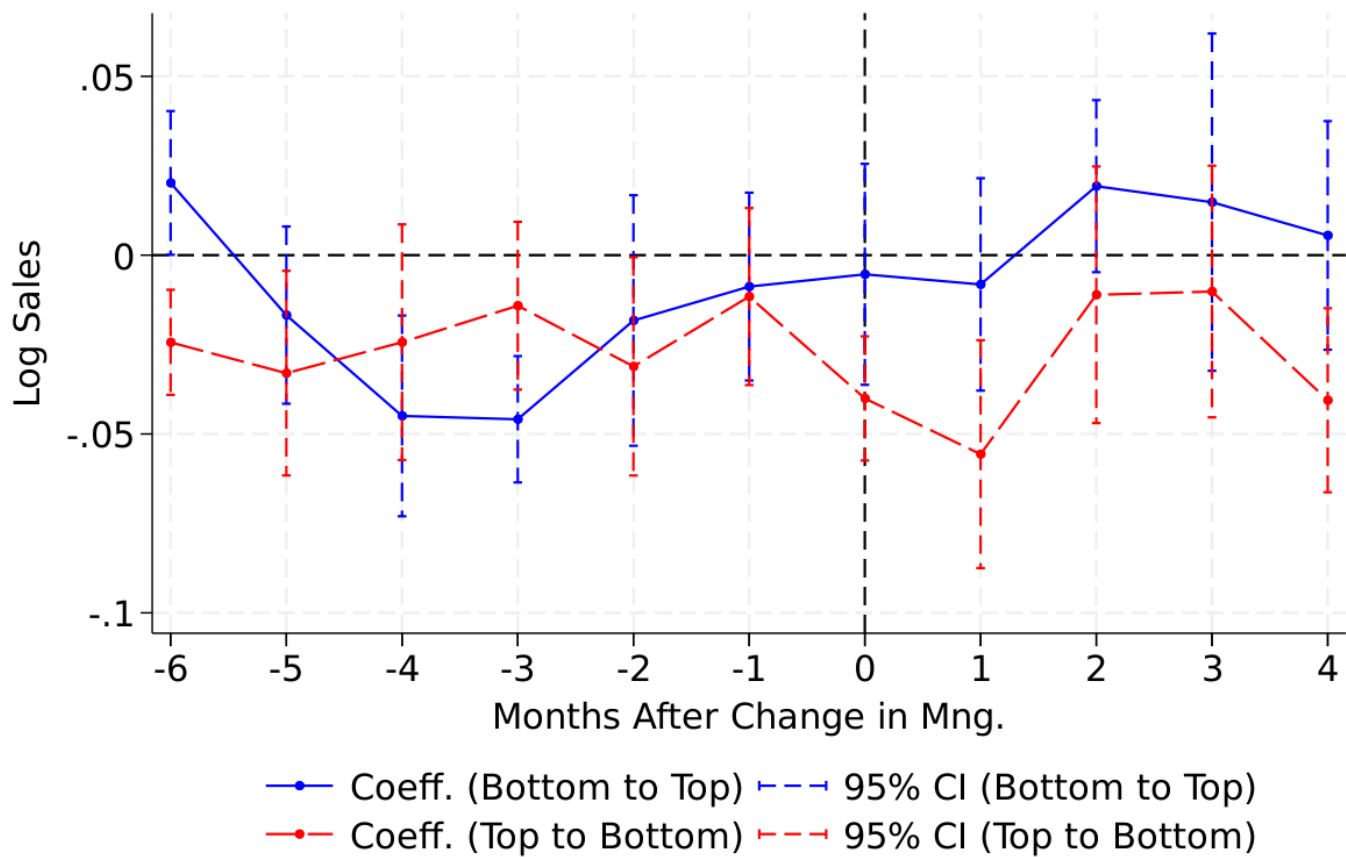
Table 3: Transition matrix of managerial moves

		Stores					
		Company A			Company B		
		Top	Middle	Bottom	Top	Middle	Bottom
Managers	Top	65	18	17	69	19	12
	Middle	22	62	16	16	63	21
	Bottom	13	21	66	14	17	69

Paper version

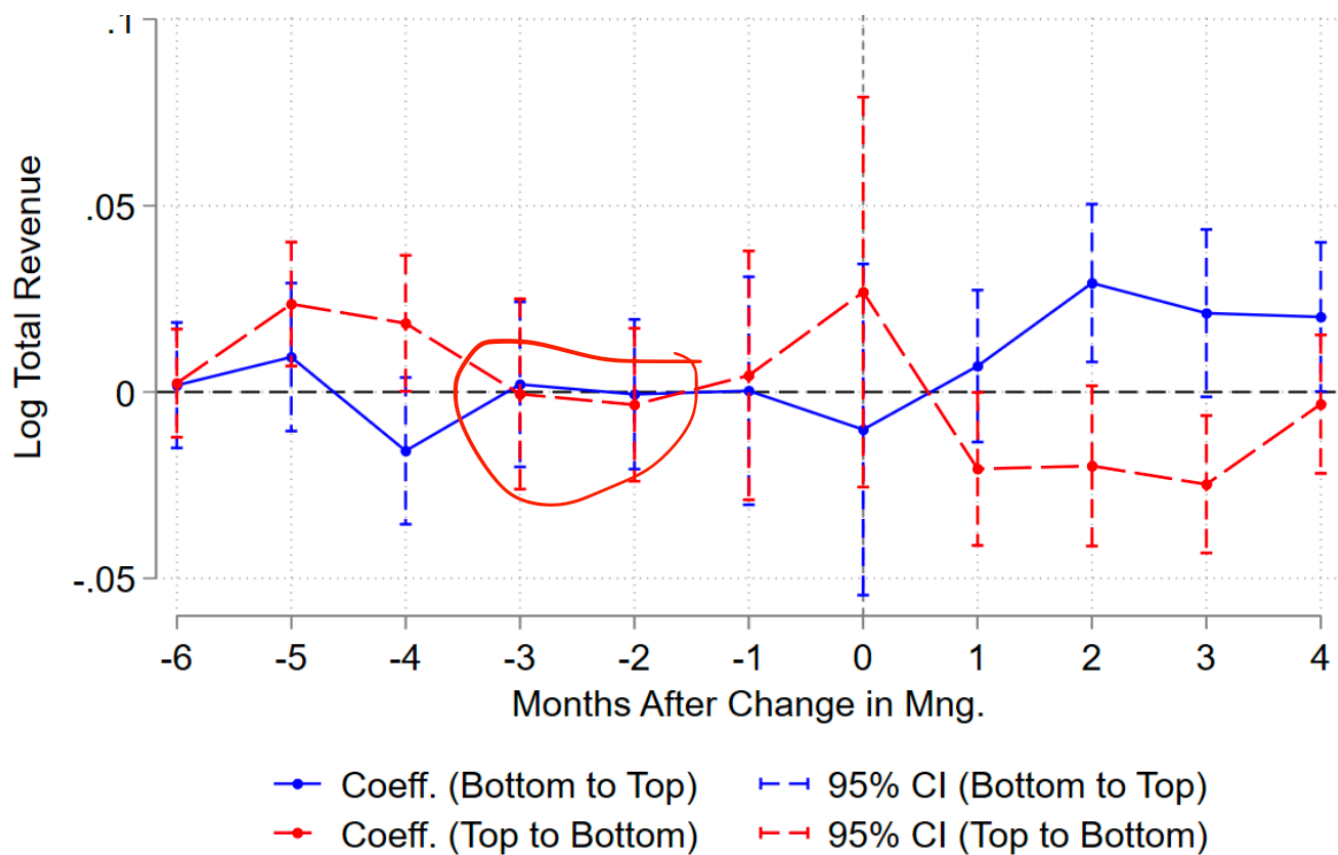
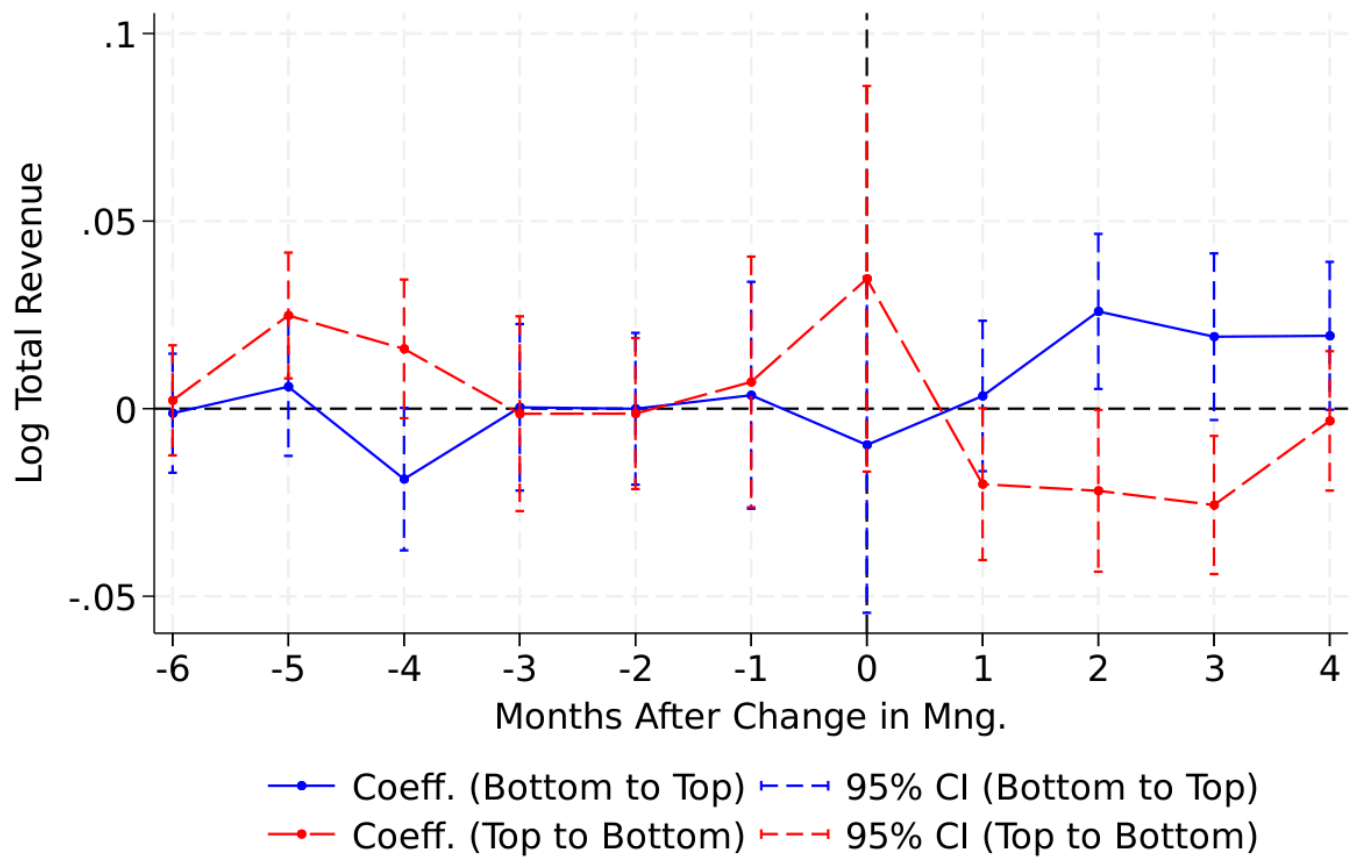
Figure 4

- Figure 8: Some of the event graphs look different. I included b) as an example below. The other graphs with divergences are c) and e).



(b) Sales (log)

- Figure 9: Some of the event graphs display discrepancies. See example (b) "Sales" below, other parts of the figure that displays discrepancies are e) and f).



(b) Sales (log)

- Table 4 not reproduced or not saved in the results folder.
- Figure A1 was not saved in the results folder.
- Figure A2 is showing discrepancies:

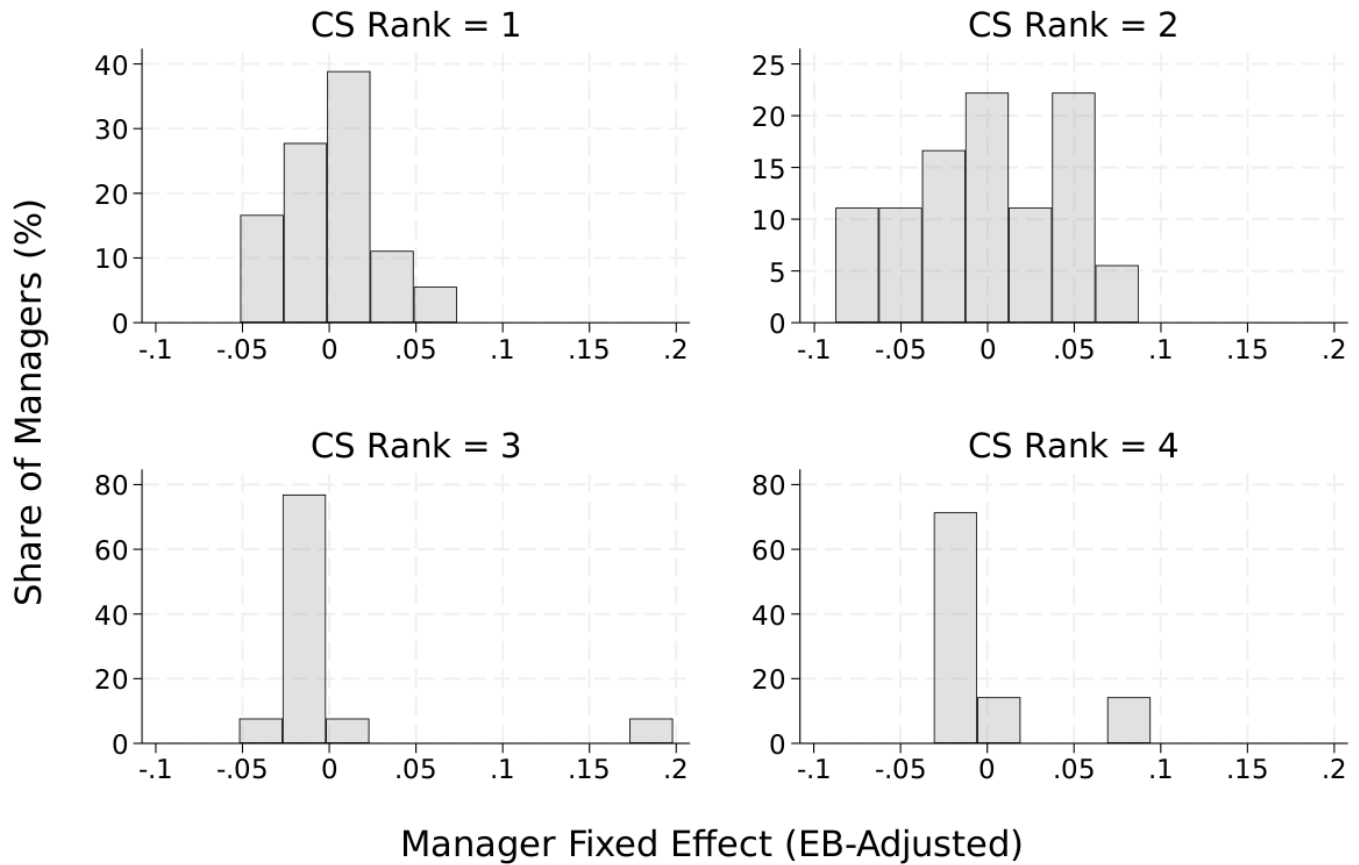
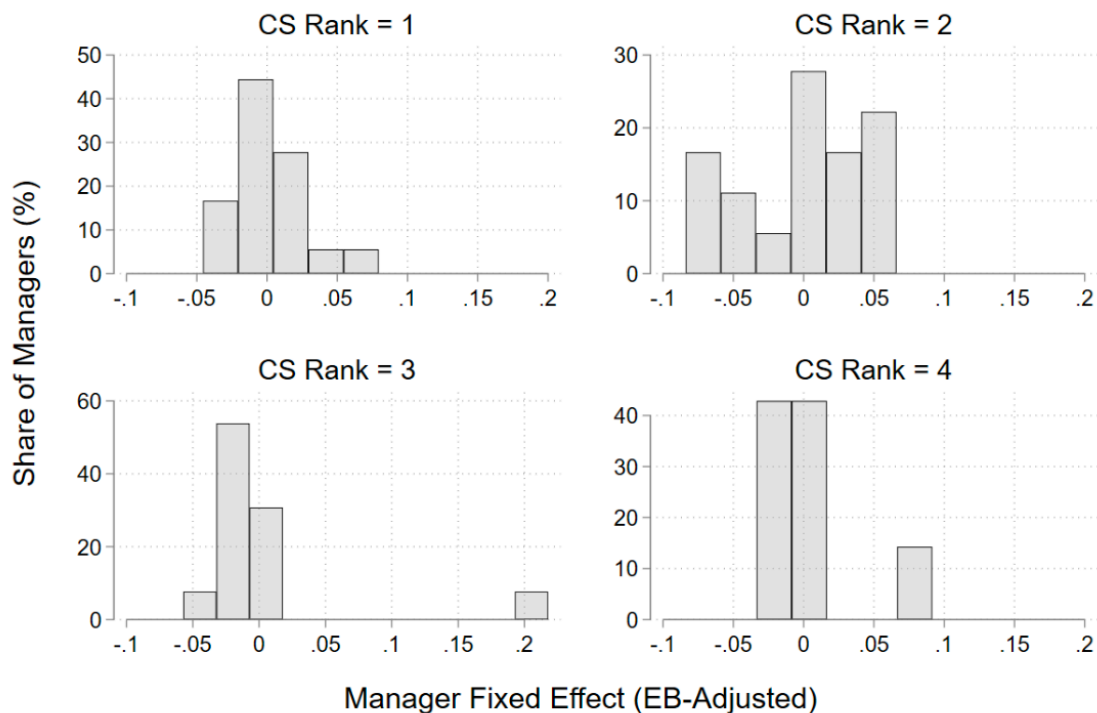
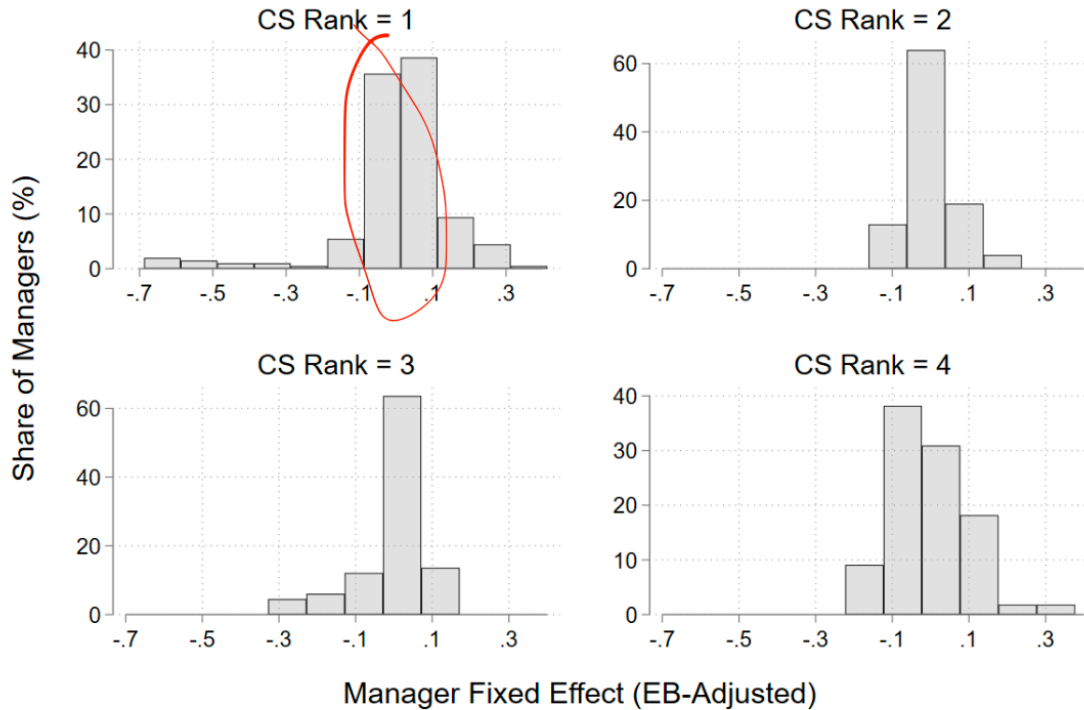
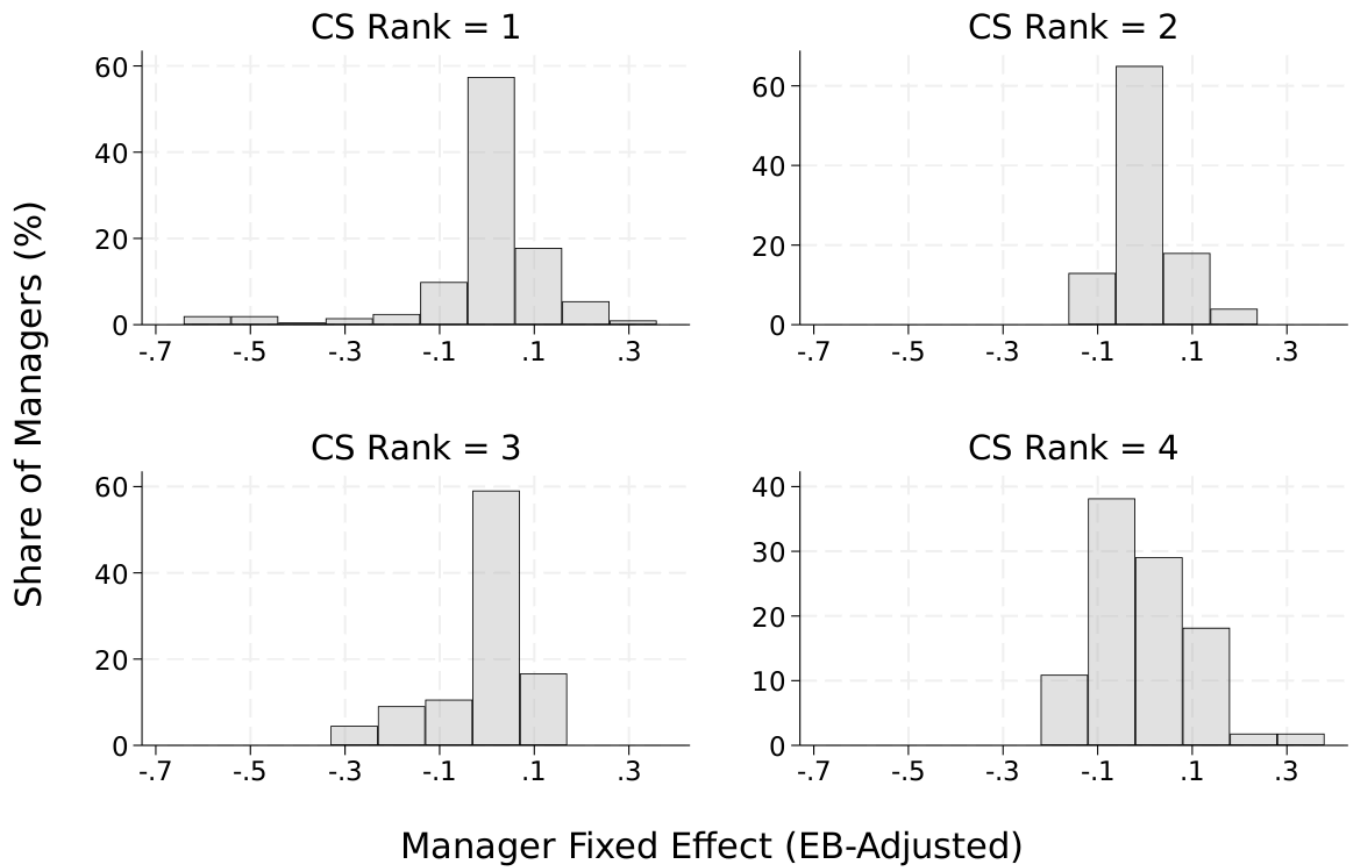


Figure A.2: Distribution of EB-Adjusted Manager Fixed Effects



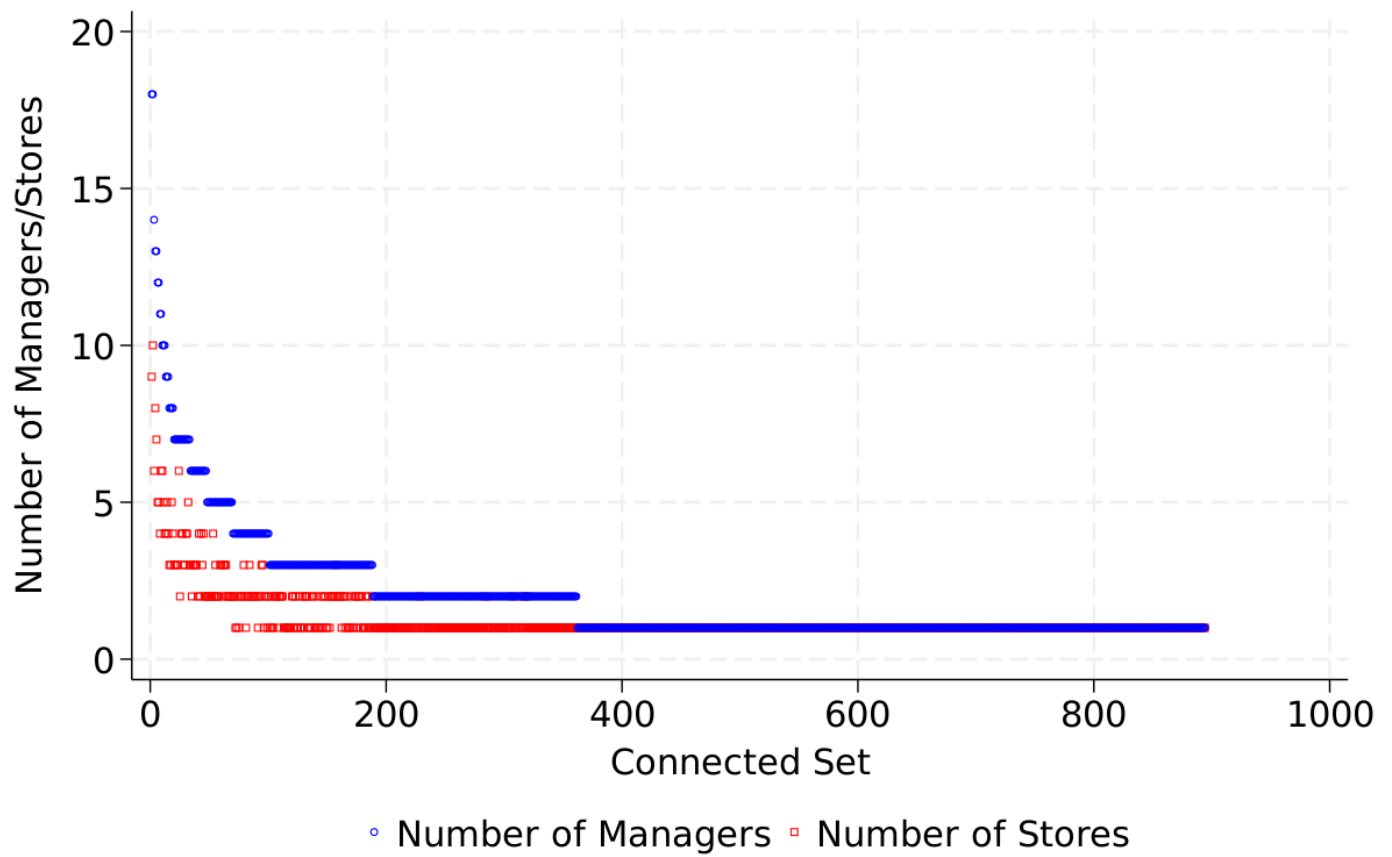
(a) Company A

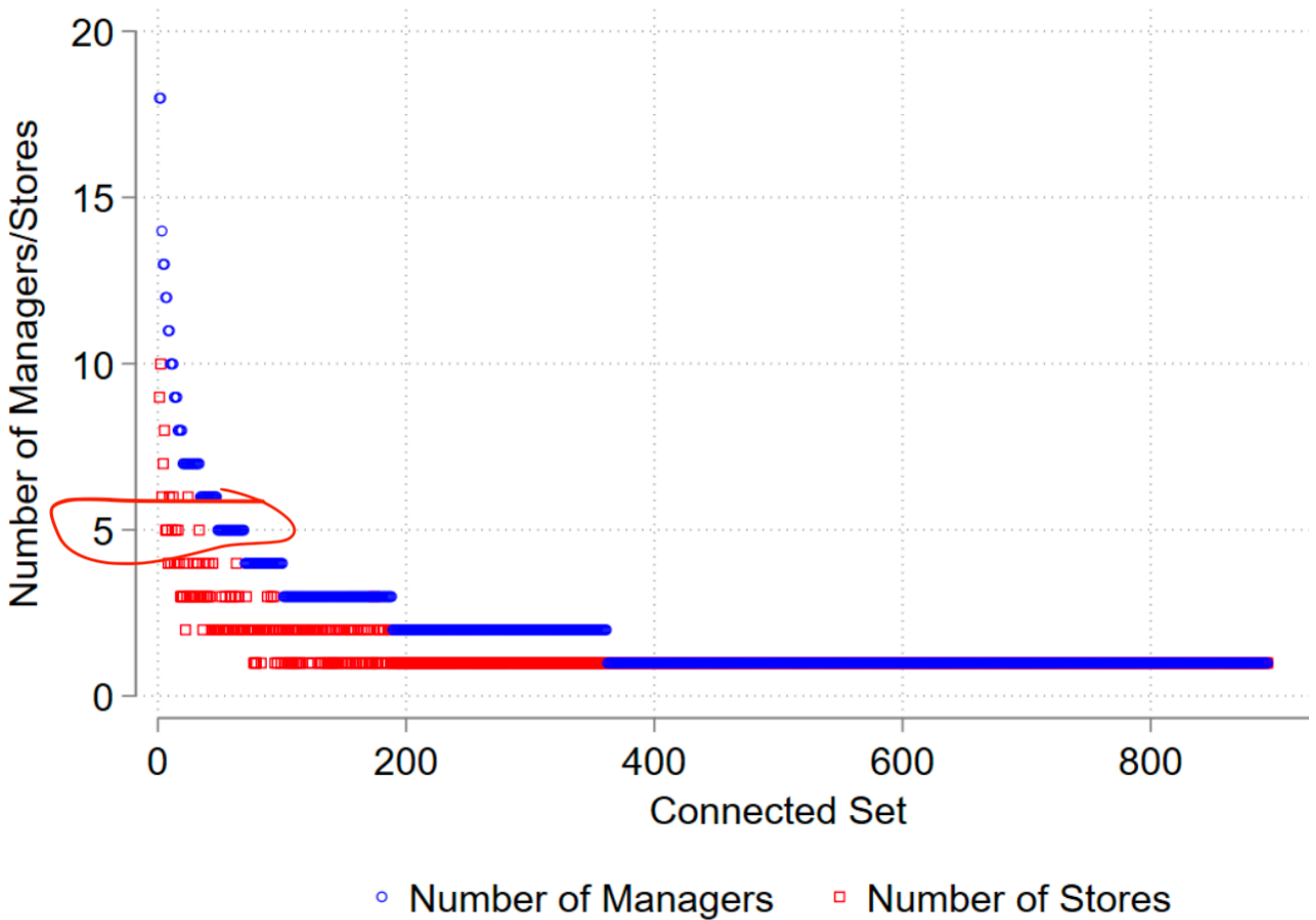


(b) Company B

Note: FE's de-meanned by connected set for plotting.

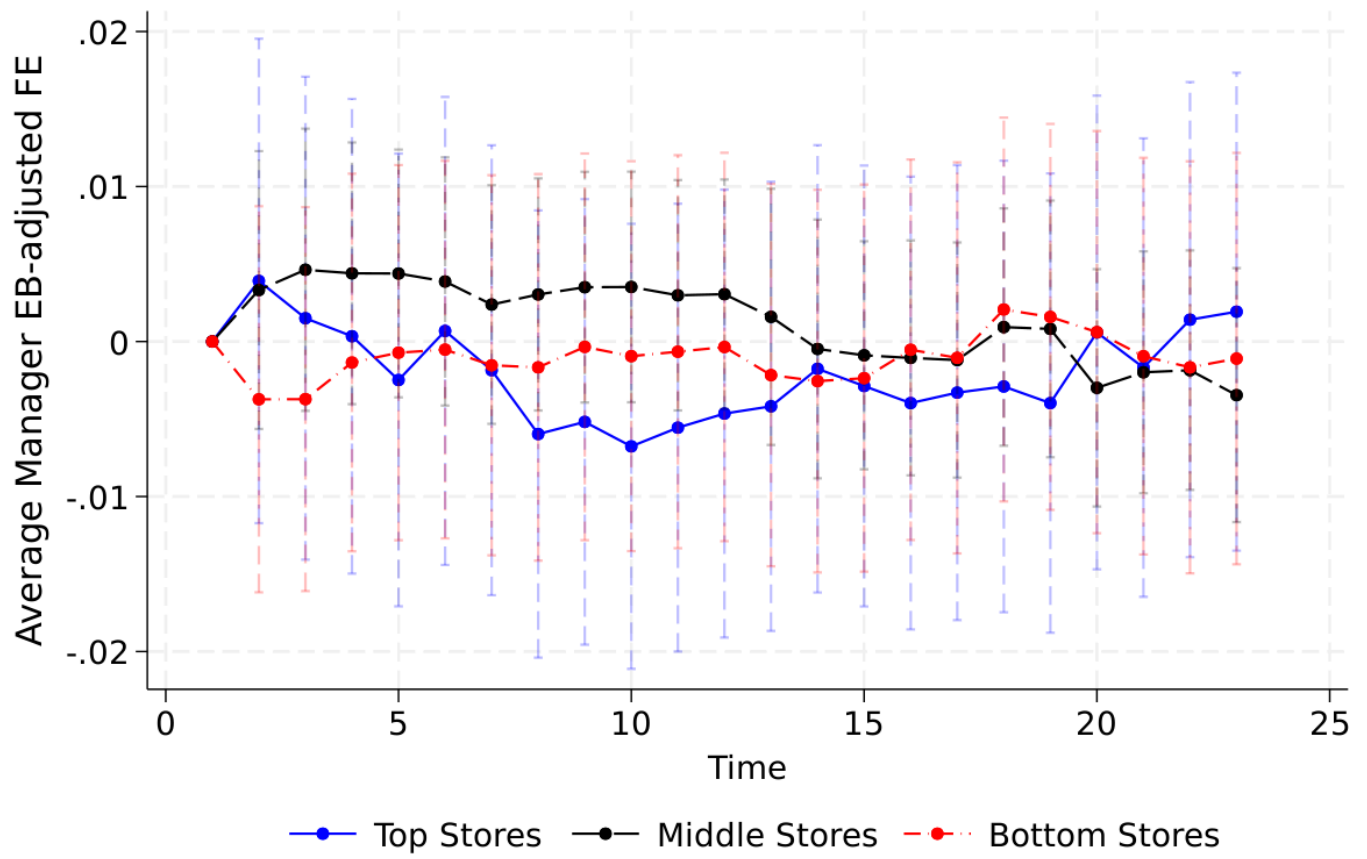
- Figure A3 showcases similar discrepancies as figure A2.
- Figure A4 Company A showcases discrepancies:

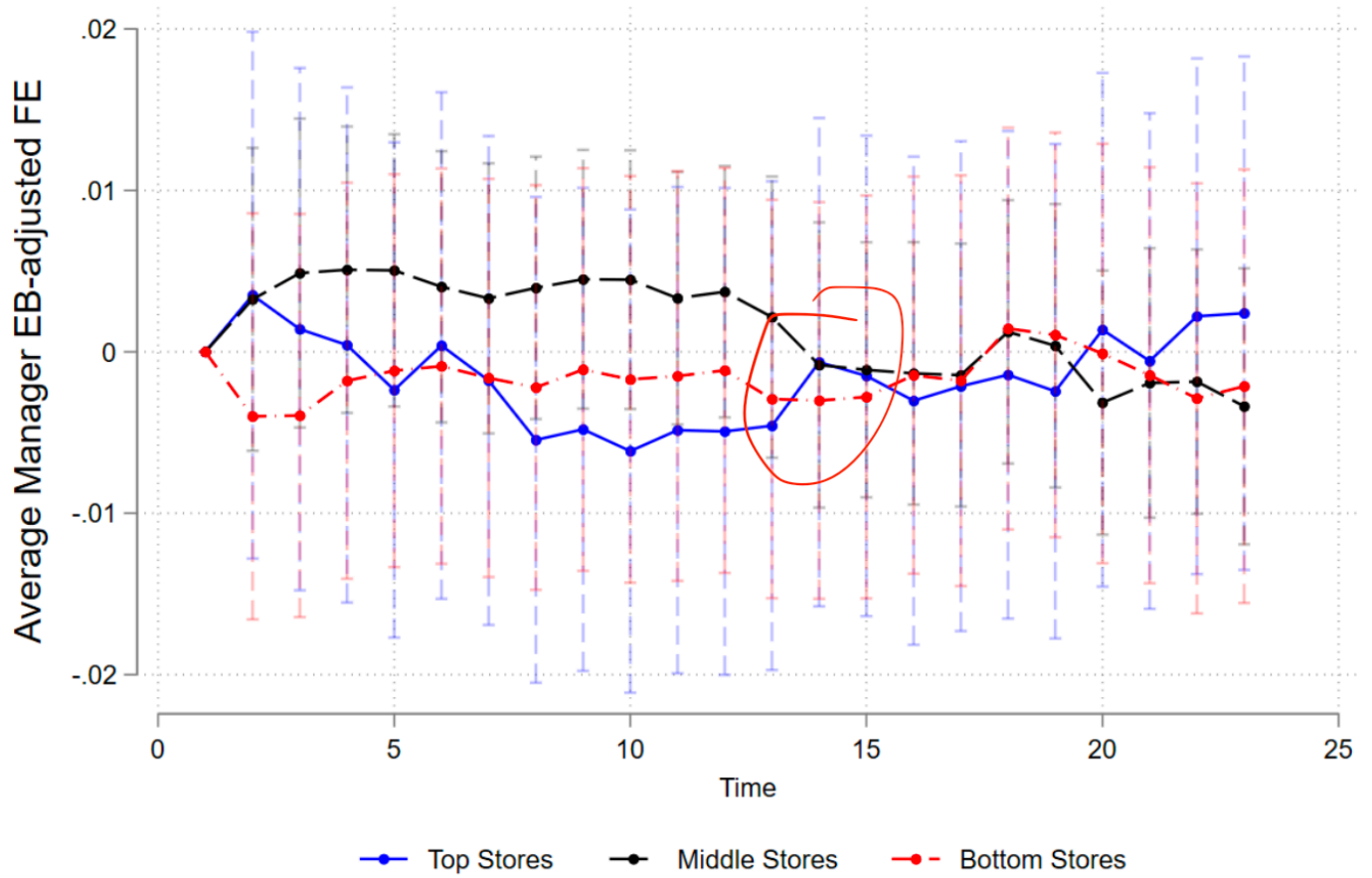




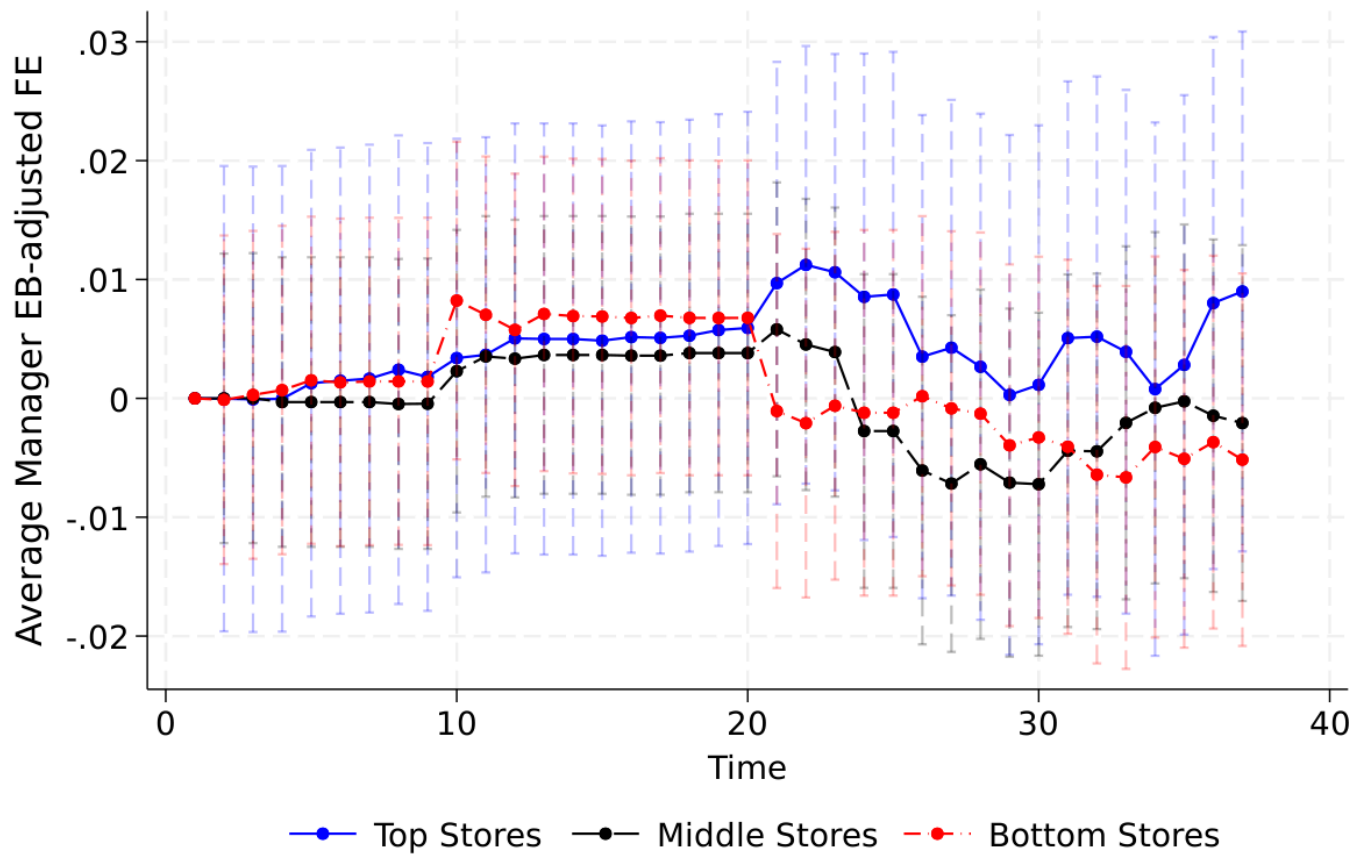
(a) Company A

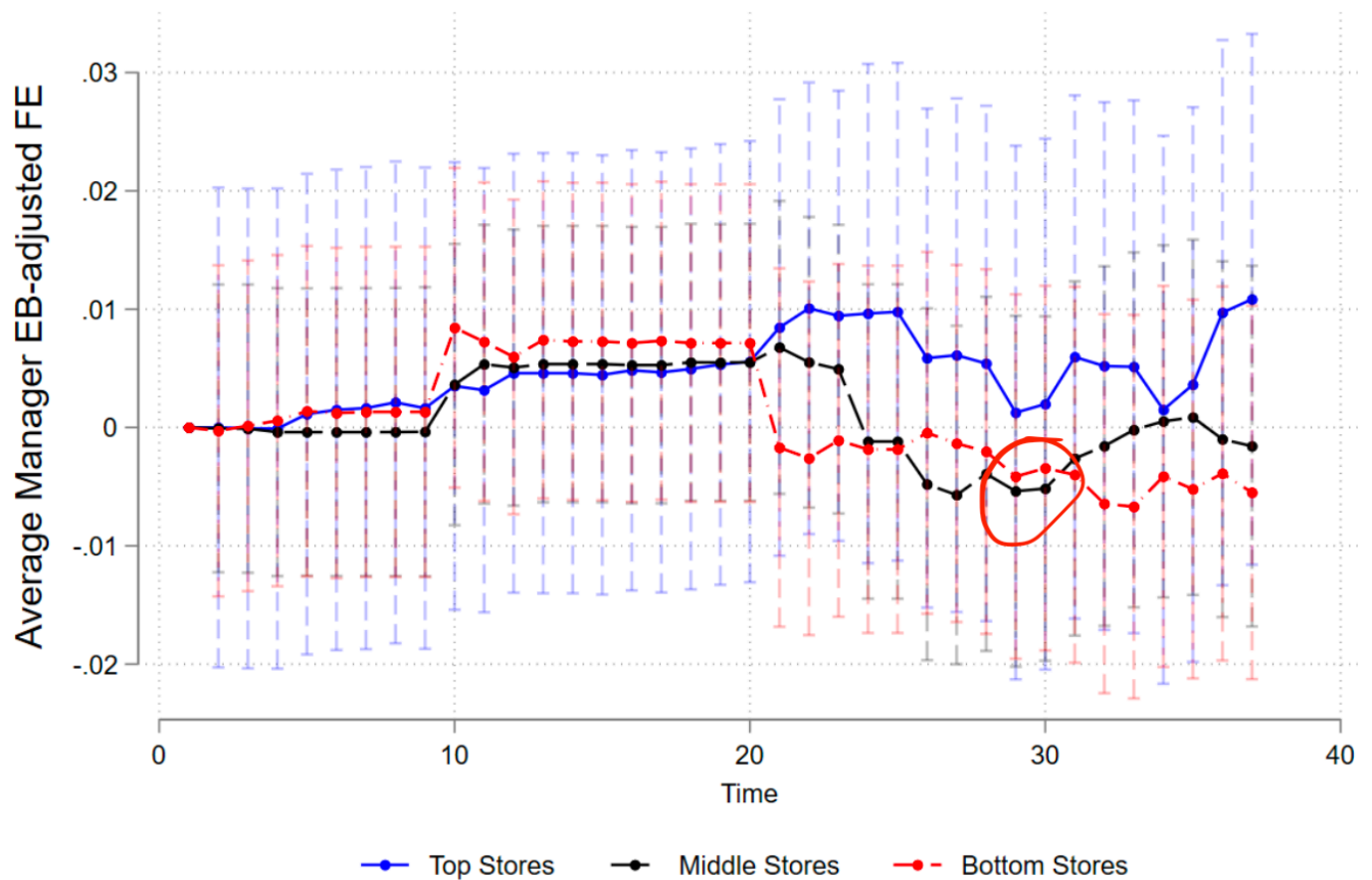
- Figure A5: Looks the same.
- Table B.1 not reproduced or not saved in the results folder.
- Table B.2 not reproduced or not saved in the results folder.
- Table C.1 not reproduced or not saved in the results folder.
- Figure C.1 not reproduced.
- Figure C.2 not reproduced.
- Figure C.3 not reproduced.
- Table D1 not reproduced or not saved in the results folder.
- Figure D1 displays discrepancies for both companies:





(a) Company A





(b) Company B

- Figure D2 looks the same.
- Figure D3 showcases same discrepancies, but the figures are based on a randomised simulation, which I could explain reasonably minor differences:

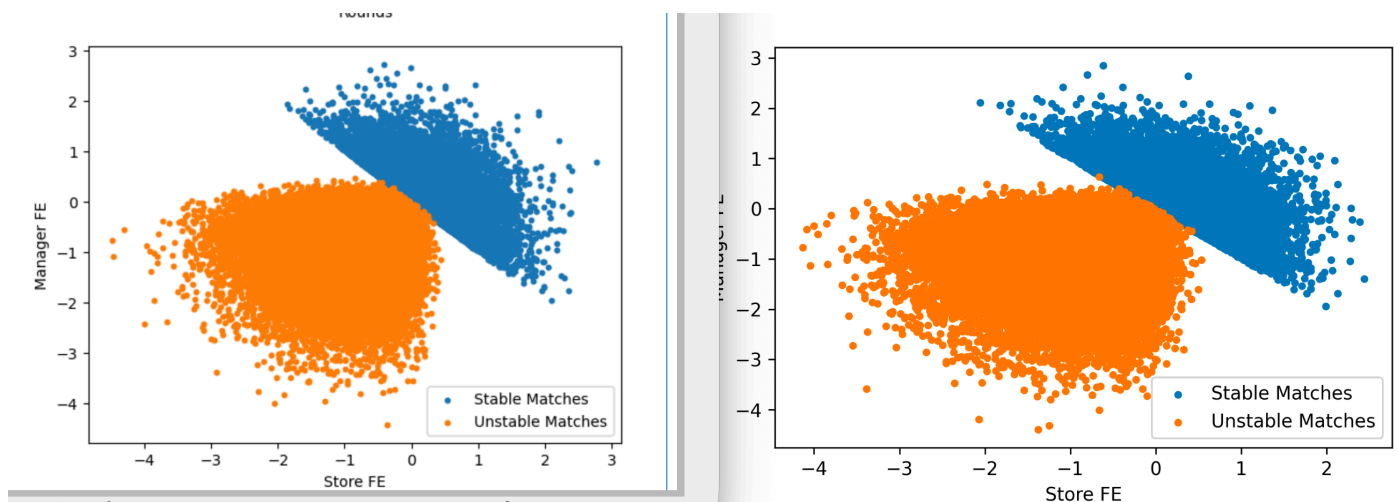
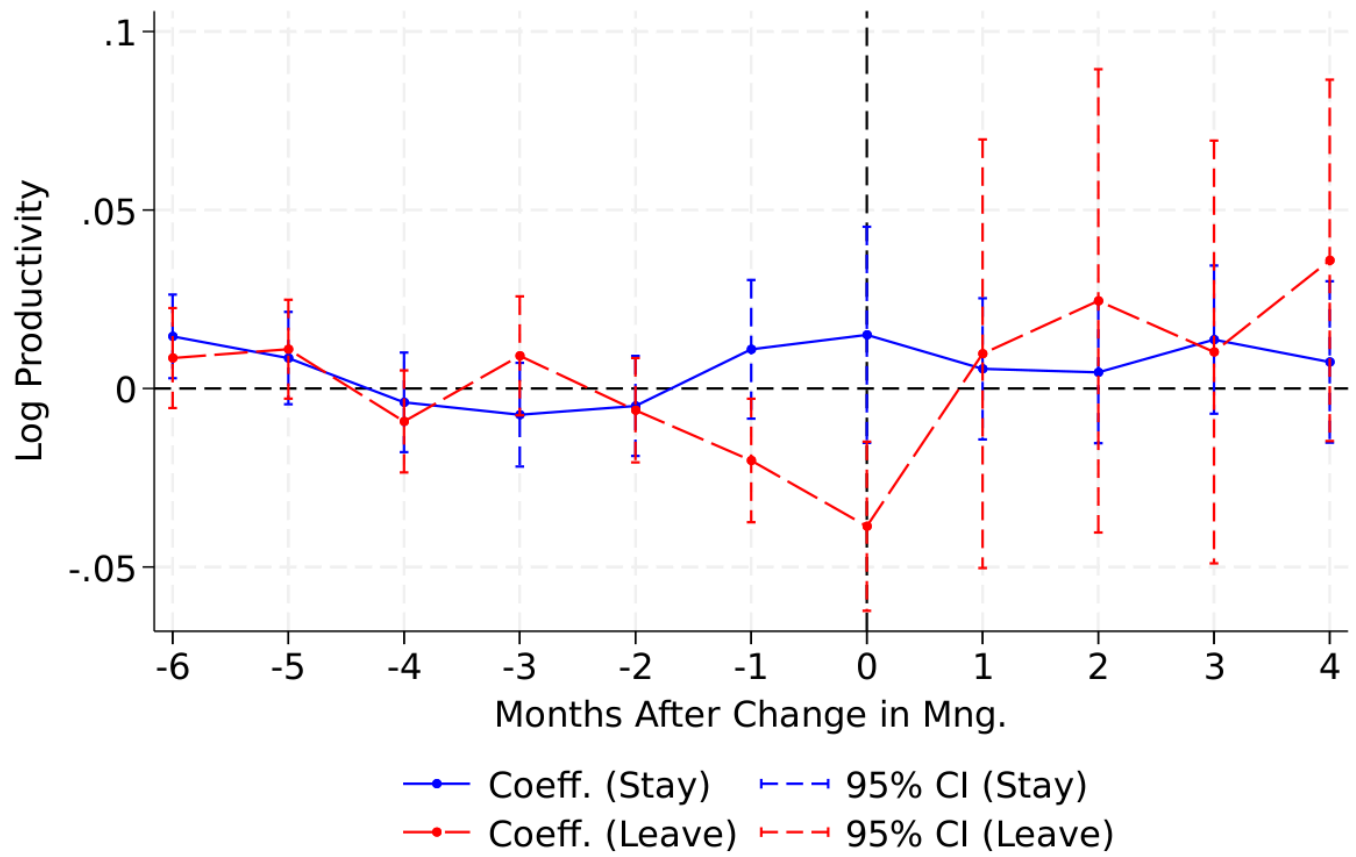
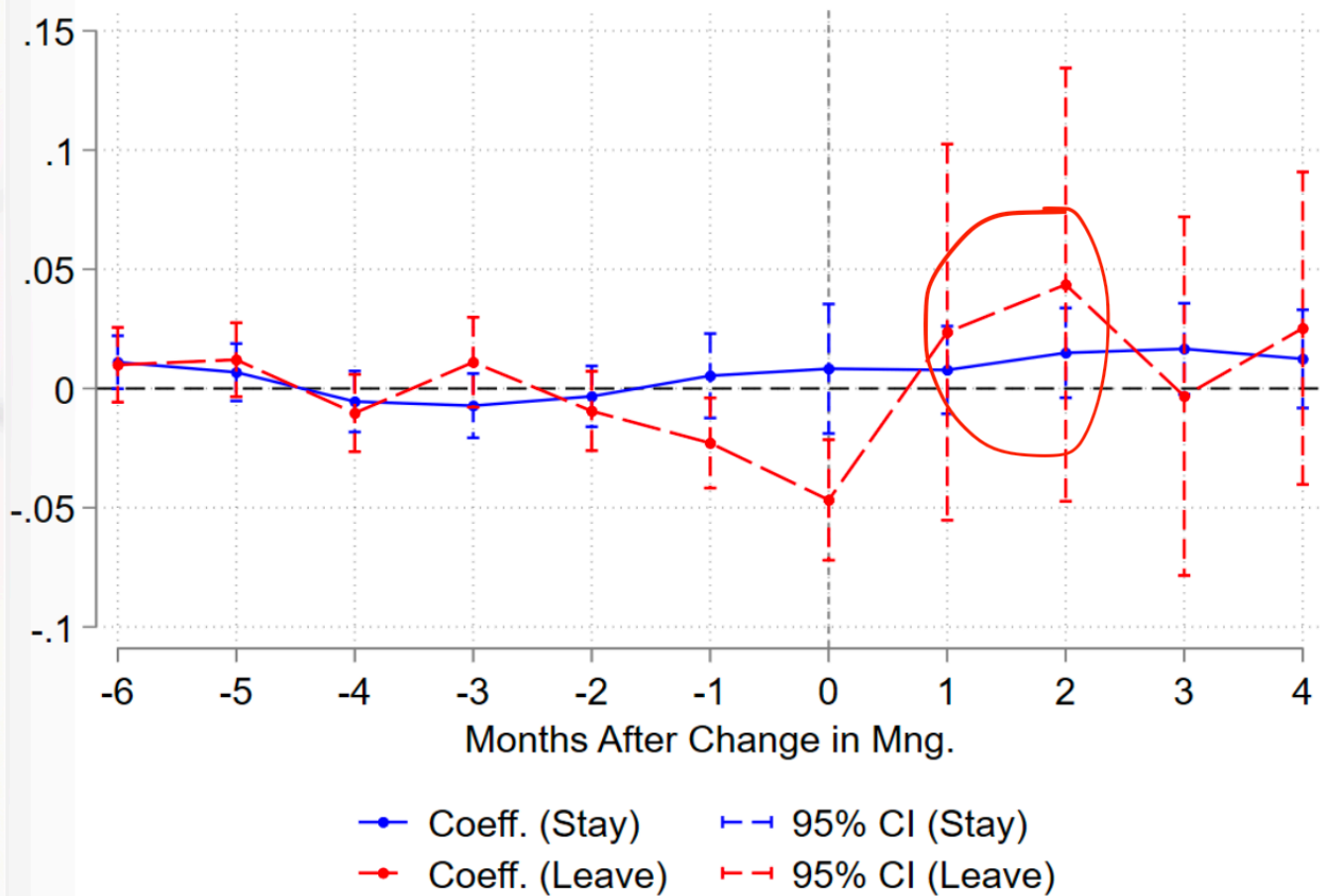


Figure D3 (b) reproduced on the left, paper version on the right

- Figure E1: looks the same.

- Figure E2: a) and c) look the same, b) and d) not reproduced.
- Figure E3 was not reproduced.
- Table F.1 was not reproduces.
- Figure G.1: looks the same.
- Figure G.2: looks the same.
- Figure G.3: looks the same.
- Figure G.4: looks the same.
- Figure G.5: looks the same.
- Figure G.6: showcases discrepancies in all graphs a) below as an example:





(a) Productivity (log)

9.4 In-Text Numbers

{REPLICATOR}: list page and line number of in-text numbers. If ambiguous, cite the surrounding text, i.e., "the rate fell to 52% of all jobs: verified".

- ☒ There are in-text numbers, but they are not identified in the code

9.5 Other Issues

- Title of the Readme needs to be adjusted.
- Details on data sources missing.
- Details on in-text numbers are missing in the ReadMe.
- Line of code where results are reproduced in the codes are not provided in the ReadMe.

- Paths are written with backslashes in the Stata do files.

10 Classification

- ☐ full reproduction
- ☐ full reproduction with minor issues
- ☒ partial reproduction (see above)
- ☐ not able to reproduce most or all of the results (reasons see above)

10.1 Reason for incomplete reproducibility

- ☐ None.
- ☒ **Discrepancy in output** (either figures or numbers in tables or text differ)
- ☒ **Bugs in code** that were fixable by the replicator (but should be fixed in the final deposit)
- ☐ **Code missing**, in particular if it prevented the replicator from completing the reproducibility check
 - ☐ **Data preparation code missing** should be checked if the code missing seems to be data preparation code
- ☒ **Code not functional** is more severe than a simple bug: it prevented the replicator from completing the reproducibility check
- ☐ **Software not available to replicator** may happen for a variety of reasons, but in particular (a) when the software is commercial, and the replicator does not have access to a licensed copy, or (b) the software is open-source, but a specific version required to conduct the reproducibility check is not available.
- ☐ **Insufficient time available to replicator** is applicable when (a) running the code would take weeks or more (b) running the code might take less time if sufficient compute resources were to be brought to bear, but no such resources can be accessed in a timely fashion (c) the replication package is very complex, and following all (manual and scripted) steps would take too long.
- ☐ **Data missing** is marked when data *should* be available, but was erroneously not provided, or is not accessible via the procedures described in the replication package
- ☐ **Data not available** is marked when data requires additional access steps, for instance purchase or application procedure.
- ☐ **Missing README** is marked if there is no README to guide the replicator, or the README is not in compliance with JPE requirements